1989 Vidmar AS/400 HP 3000 with 2024 Metadata

I. IN WHAT WAY IS ABERDEEN GROUP QUALIFIED TO CONSULTTO STANLEY- VIDMAR ON MATTERS CONCERNING THE IBM AS/400 AND THE HEWLETT- PACKARD 3000 MARKETPLACE?

ABERDEEN GROUP QUALIFICATIONS

Aberdeen Group is a Boston-based, internationally recognized computer consulting and research organization.

Aberdeen Group performs specific projects for a select group of domestic and international clients requiring strategic and tactical advice and pragmatic, experience-based action plans. Assignments range from strategic audits and implementation plans to corporate and product positioning to in-depth market, acquisition/divesture, and feasibility studies to benchmarking verification.

Aberdeen's principals -- Charles T. Casale, Peter S. Kastner, John R. Logan, and Thomas H. Willmott -- are recognized industry figures with over 80 years of combined high-tech industry and financial community experience among them. They are quoted extensively in industry and business publications.

In addition to client-related research and consulting, Aberdeen publishes several periodicals summarizing its analysis and research findings:

Executive Viewpoint -- a strategic industry review analyzing major trends affecting suppliers and users. Each issue focuses on a specific topic. One recent issue showed how subtle changes in user perceptions are causing major shifts in buying practices, with serious import for many vendors. Another Executive Viewpoint focused on how an increasing cadre of outside observers are distorting buyers' and vendors' communications channels and impeding sales.

Product Viewpoint: Technology Viewpoint; Market Viewpoint. This series of product, technology and market analyses are issued in response to new developments and review the implications, opportunities, and consequences. Viewpoints are published monthly for Aberdeen clients. Although they are not available on a subscription basis, Aberdeen has reprinted numerous issues at client request for reasonable fees. A recent, popular Product Viewpoint, AS-400: Back to the Future, focused on IBM's newest and widely discussed midrange offering -- and has been credited by the industry as the first research report that provided users with a framework for analyzing the performance of the AS/400 in relationship to both comparable Hewlett-Packard and Digital systems.

Research Studies. In-depth market studies of topical interest are sold individually. Current reports include Conflicting Trends in Computational Chemistry, and Transaction Processing -- Into the 1990s.

BIOGRAPHIES CHARLES T. CASALE

Charles T. Casale is President and co-founder of Aberdeen Group, Inc. Mr. Casale is a business executive with a diversified 32-year background in corporate development, strategy, finance, marketing, communications, litigation support and international operations. He is also a securities analyst who has followed IBM and Hewlett- Packard for over twenty years.

Mr. Casale's operational experience includes being a founding officer of three high-technology companies (Taxon; Encore Computer; DQ Securities), architect of a major computer system (the CDC 3600) and director of corporate development at Prime Computer. He began his career as a large-scale computer systems architect after earning a BSEE. He is a Chartered Financial Analyst, and holds a basic computer patent.

Mr. Casale's professional affiliations include the ACM; the IEEE, where he is a senior member; the National Investor Relations Institute, where he has achieved national honors; The Institute for Chartered Financial Analysts; and the New York Society of Securities Analysts.

JOHN R. LOGAN

John R. Logan is executive vice president and co-founder of Aberdeen Group, Inc. Mr. Logan is one of the nation's leading analysts of trends in the computer industry. Before joining Aberdeen, he was director of mid-range computer research at the Yankee Group where he followed HP and IBM products as well as co-authored the report *The Future of Transaction Processing*.

Mr. Logan previously was a vice-president and co-founder of a software publishing firm, concentrating on the resale of major applications packages. In addition, he joined Prime Computer in 1980 to establish a third-party software program. He also headed a task force for the top management to determine future market and technological trends facing the company. Upon completion of his undergraduate studies, Mr. Logan was a corporate industrial engineer with American Greetings Corporation, responsible for manufacturing and distribution operations.

Mr. Logan earned a BSE with a major in Industrial Engineering from the University of Michigan and an MBA from Harvard University.

PETER S. KASTNER

Peter S. Kastner is vice president of Aberdeen Group, Inc. and is general manager of Aberdeen Transaction Services, an Aberdeen unit providing industry analysis and consulting in the fast-growing transaction processing market.

Mr. Kastner has an industry perspective gained over twenty years of working with end users in designing and implementing complex computer systems, and with major equipment manufacturers in developing marketing and sales strategies. He is widely regarded as a leading expert in transaction processing. He manages the Aberdeen Group performance benchmark activities, and serves as an expert witness on the subject of capacity planning and performance measurement.

Mr. Kastner has held marketing and management positions with Digital Equipment, Prime Computer, and Stratus Computer. He has been a consultant with Arthur D. Little where for 9 years he led numerous system design and implementation projects for financial, government and manufacturing clients. He was also responsible for the A.D. Little OEM business with HP 3000, IBM System/3X, DEC, and DG.

THOMAS H. WILLMOTT

Thomas H. Willmott is vice president of Aberdeen Group, Inc. and is general manager of its consulting operations. Prior to joining Aberdeen, Mr. Willmott was Executive Vice President of International Data Corporation, a Massachusetts-based computer and communications research and consulting firm. During his eight years with IDC, Mr. Willmott served in various capacities, most recently managing worldwide research, all U.S. technology research products and services, and IDC' Asian operations. From 1987 to 1988, Mr. Willmott ran IDC's European Research Center in London.

Mr. Willmott is a frequent conference speaker and is author of Software Solutions for the IBM PC (Prentice-Hall 1983). He has a masters degree in education and graduated from Williams College with a concentration is Russian Area Studies.

II. COMPARE THE QUALIFICATIONS OF ABERDEEN GROUP TO THOSE OF THE GARTNER GROUP AND OTHER RESEARCH FIRMS. IS THE ABERDEEN GROUP MORE EXPERIENCED OR COMPETENT IN THE MIDRANGE ARENA?

SUMMARY

The roots of Aberdeen Group originate from the Route 128 midrange computer industry center. As a result, Aberdeen's perspective on the midrange marketplace is considered by the press (Wall Street Journal, Business Week, Computer World, MIS Week) and suppliers (Data General, Stratus, Prime, Hewlett-Packard, DEC, Unisys, IBM, Wang and numerous software

enterprises) second to none.

FINDINGS

- The principal business of Gartner Group is subscription market research. Gartner Group does not generally provide direct consulting services focussed on specific customer requirements beyond what can be answered in a telephone call. A substantial portion of Gartner Group research is derived from supplier sources.

- The principal business of Aberdeen Group is custom consulting to user and supplier clients, while a minor portion is subscription market research. Aberdeen Group usually provides consulting on specific client issues. *A substantial portion of Aberdeen research is derived from user interviews -- not supplier marketing hype.*

- Aberdeen is primarily concerned with user business issues while Gartner Group focusses on product comparisons.

- Aberdeen principals have diverse business and technology backgrounds measured in decades each where the research emphasis is on the inter-relationships of technologies with changing operational requirements dictated by an increasingly competitive environment in which our clients operate. Gartner's research is done by analysts with a narrow product focus dictated by the subscription service module each is responsible for.

- No single Aberdeen client is large enough to influence our research findings. Neither IBM nor Hewlett-Packard constitute a substantial percentage of our revenues. As a result, we are not an extension of either's public relations department.

- Forrester Research is in the newsletter and seminar business, not consulting. As a result, Aberdeen and Forrester frequently recommend each other's services.

- International Data Corporation (IDC) is well known as the official numbers keeper of the industry. Aberdeen frequently uses IDC numbers to conduct analysis of trends in the industry.

- The consulting services of accounting firms are in the business of installing systems and providing programming services. Aberdeen is not in the contract programming business, but stays more current with user and technology trends to recommend information strategies for our clients to implement in the way most suitable for their style of management.

IMPLICATION

- Stanley-Vidmar's computer selection problem is best suited to a custom consulting project where the consultant can review the current installation and business objectives of Stanley-Vidmar -- not just compare an AS/400 versus a Hewlett-Packard system in a vacuum.

CONCLUSIONS

- Aberdeen is in the business of solving specific client problems through custom consulting. Gartner Group's business is directed at written commentary of interest to a wide audience, leaving the reader to make sort out the issues for his specific problem.

- Aberdeen Group is publicly recognized for its objective, unique, pragmatic, user-oriented focus on solving technology-related business problems.

- Aberdeen Group is recognized in the industry as being uniquely different from the Gartner Group and most other consulting firms.

III. GIVEN THE CURRENT INFORMATION SYSTEMS ENVIRONMENT, WHICH UPGRADE PATH (AS/400 OR HEWLETT-PACKARD) IS BETTER FOR STANLEY-VIDMAR? WHY?

HP PRECISION ARCHITECTURE FINDINGS

- Analysis of performance measurement data collected for June and July unequivocally indicate CPU saturation as the prime culprit for poor response time at Stanley-Vidmar. Other metrics -- memory, process, and I/O -- are within acceptable bounds.

- We believe a 100-user production system at Stanley-Vidmar requires a mid-point model in the Hewlett-Packard Precision Architecture (HPPA) product line. Stanley-Vidmar's requirements for future growth appear to be more than adequately covered by announced HPPA products.

- Stanley-Vidmar has a significant investment in learning how to use HP systems to meet its operational objectives. The tangible aspects are programmer skills, operations, tuning, and application software. The intangible assets are in training and "job knowledge", and in clearly understanding how to get from today's application environment to tomorrow's.

- HP is the low risk approach. The IS staff are already expert with Hewlett-Packard's MPE operating system. The HP manufacturing software package comes with HP's implied promise to make it work at Stanley-Vidmar. HP Precision Architecture and MPE/XL are mature products which demonstrate high reliability and have been highly praised by users.

- Existing "low priority" programs can run indefinitely in emulation mode (MPE V) without requiring conversion and testing to the HPPA MPE XL operating system, allowing the MIS staff to concentrate on high-value renovation and new applications.

IBM AS/400 FINDINGS

- The AS/400 is not yet a mature product. Adequate tape drives were only recently announced. Operating software in particular lacks maturity. Modern decision support applications are missing due to the lack of a C compiler. Program size is limited to 1 MB. Users complain of numerous program fix tapes, making it difficult to run production and keep up with O/S patches.

- OS/400 uses a single-level storage system that was advanced when it was first pioneered in the 1970s. While capable of yielding high I/O performance, user databases are much more likely to require lengthy restores after a system failure than conventional midrange computer storage approaches. Crash recovery procedures are crucial for mission-critical applications such as those used by Stanley-Vidmar.

Hewlett-Packard uses a more conventional approach to storage systems augmented by Reduced Instruction Set Computing (RISC) features that improve I/O performance. Crash recovery is not as involved as with the AS/400.

- IBM RAMP-C benchmarks do not use Journaling and Checksum software which would speed up system recovery after a crash. As a result, the high-performance shown from such tests is illusory -- real-world production systems will have much lower performance. However, Aberdeen strongly recommends Journaling and Checksum even though these features can reduce throughput by up to 60%.

- Users and implementers of the AS/400 consistently complain that realized performance is approximately 30% less than that projected by IBM's AS/400 analysis tools. ADM, Inc., a Cheshire, Connecticut System/3X and AS/400 consulting firm and AS/400 Value Added Distributer has publicly told IBM that the IBM AS/400 sizing methods lead to underconfigured systems.

- The fact that some existing Stanley-Vidmar programs originated in System/3X-based MacPac is not necessarily helpful. We believe this code will require about as much conversion effort as other program code.

- An AS/400 is a high risk approach for Stanley-Vidmar. The MIS staff will require extensive training and preparation prior to conversion. The Business Planning Control System (BPCS) manufacturing package is supported by local distributors, a tier removed from the author, System Software Associates, Inc. in Chicago. The sales analysis system must be developed from scratch using outside consultants. *Running parallel and cutover with two different computing systems will tax the limited resources of the MIS programming and operations staff,* leaving little time to manage the outside software contractors.

- Given the high risks of the AS/400 approach, cost escalation of implementing AS/400 at

Stanley-Vidmar is foreseeable in three major areas:
. intangible costs in getting the MIS staff up to speed on a new computer are

typically underestimated.
. There have been AS/400 persistent user complaints about poor production

performance, implying that more hardware is needed.
. The new sales administration system project may require more consultant

effort than estimated.

IMPLICATIONS

- Leaving its current Hewlett-Packard operating environment will increase costs and set back the operational effectiveness of Stanley-Vidmar's MIS department.

- Stanley-Vidmar should only move to a non-Hewlett-Packard environment if substantial cost savings and/or increased operational effectiveness can be shown. The AS/400 has not demonstrated in real-world production environments superiority to the Hewlett-Packard solution.

- An upgrade to the more powerful Hewlett-Packard Precision Architecture will benefit Stanley-Vidmar users immediately by reducing response times while running existing applications.

- An AS/400 conversion will likely exceed a year in duration before all the pieces are in place due to the large amount of converted code and the new sales analysis system design and implementation.

- Upgrading to a more powerful HPPA system implies the extension of current in-house expertise and the use of local Stanley-Vidmar resources to ensure operational success.

- An AS/400 decision implies new local expertise will be required. Stanley corporate MIS will not be actively involved in the implementation at Stanley-Vidmar. Thus, Stanley-Vidmar must retain outside AS/400 expertise and invest in MIS training and development.
- Historically, the relative lack of computing power in the System/3X kept that system at numerous, smaller, typically unsophisticated-in-data processing manufacturers. The HP 3000 was designed for higher performance users in larger manufacturing operations. The current implication is that the AS/400 now has the computing power to handle larger manufacturers which the System/3X did not. But we believe that the *needs and requirements* of larger manufacturers may not be well understood by the majority of the AS/400 sales force or IBM's AS/400 software business partners.

CONCLUSIONS

- For Stanley-Vidmar to convert their current manufacturing applications to an AS/400

will absolutely involve more effort, risk, and total cost than an HP upgrade. More to the point, there are no apparent significant operational effectiveness, economic or technological benefits to Stanley-Vidmar in converting to an AS/400 system.

- An HP 3000 can readily run today's software much faster, solving the response time problem immediately. The existing in-house HP skills are a valuable intangible asset which should speed the efficient use of the new machine for existing and new applications.

- Stanley-Vidmar has significant leverage with the HP local sales office as a potential HPPA and manufacturing software customer. Stanley corporate volume discounts for the AS/ 400 can be used to negotiate with HP.

- Aberdeen Group recommends that Stanley-Vidmar follow an upgrade approach based on Hewlett-Packard Precision Architecture rather than an AS/400 conversion.

IV. COMPARE THE VARIOUS COMPUTERS IN THE HP 3000 PRODUCT LINE IN TERMS OF THROUGHPUT CAPACITIES.

INTRODUCTION TO BENCHMARKS

It is usually impractical during a computer selection to test all existing applications on the proposed new computer because of the time, effort and cost involved. However, buyers need to be able to ascertain the projected performance and capacity of the proposed new computer(s). For many years, buyers and sellers have relied on *benchmarks* to predict performance and capacity. A benchmark is a suite of known software run on a proposed hardware configuration for the purpose of determining the computer's capacity and performance. Thus, a benchmark is analogous to EPA milage ratings on automobiles: known procedure; useful in product-to-product comparisons; your milage (performance) may vary.

The **DebitCredit Benchmark** has evolved during the 1980s as the de facto standard for measuring performance in online database/data communications applications. DebitCredit's roots are in a branch banking application where a customer's account, teller total, branch total, and history databases are updated. The DebitCredit benchmark is elegantly simple to understand, but the benchmark intensely stresses the *whole hardware/software computer system*. Even though DebitCredit has banking roots, it has been used across industries to characterize commercial, online performance.

The performance measurement resulting from DebitCredit is transaction throughput measured in completed transactions per second (TPS). The throughput is measured at a point where 95% of all transactions complete in one second or less.

Because computer suppliers have run the DebitCredit benchmark under a variety of conditions, Aberdeen Group has normalized the benchmark's results so the results can be used to compare computers on a supplier-by-supplier basis.

FINDINGS

The online transaction processing performance of the HP 3000 Models 48, 70, 925LX, 935, 950, 955 and 960 is depicted in Figure 1.

The basis for Aberdeen's performance ranking is our proprietary normalization of the DebitCredit benchmark. We base our conclusions on:
- HP's version of DebitCredit (as documented in *HP 3000 Debit Credit Report*, December 1988, which is summarized in HP's Performance Brief entitled *OLTP DebitCredit Benchmark -- HP 3000 MPE XL 1.1*)

- - HP's published performance of the Models 48 and 70 relative to HPPA models
- - Discussions with the independent laboratory which conducted the DebitCredit tests under contract to HP

- Aberdeen's extensive experience in analyzing OLTP benchmarks.

IMPLICATIONS

- - The HPPA models offer a substantial upward growth path from the Models 48 and 70.
- - Aberdeen expects Allbase SQL performance to improve by 50% during 1990.

- \- Aberdeen expects TurboImage performance on HPPA to improve by 20% in 1990.
- \- While the 935 demonstrated excellent price/performance, it is not upgradable to the 950 and larger machines.

CONCLUSIONS

- HP can demonstrate more than 2.5 times the performance of Stanley-Vidmar's model 70 on the HPPA model 950, with a growth path into the 955 and 960.

The online transaction processing performance of the HP 3000 Models 48, 70, 925LX, 935, 950, 955 and 960 is depicted in Figure 1.

The vertical axis shows DebitCredit throughput in transactions per second where 95% of all transactions complete in a second or less. The horizontal axis shows the Hewlett-Packard models operating under the TurboImage database and the Allbase SQL relational database product. Note that the Models 48 and 70 do not support Allbase SQL.

V. COMPARE THE VARIOUS COMPUTERS IN THE AS/400 PRODUCT LINE WITH THE HP 3000 48 AND 70.

FINDINGS

The online transaction processing performance of the AS/400 models B10, B20, B35, B45, B50, B60 and B70 is depicted in Figure 2. For comparison purposes, we have included the HP

3000 48 and 70 computers now installed at Stanley-Vidmar.

The basis for Aberdeen's performance ranking is our proprietary normalization of the DebitCredit benchmark. We base our conclusions on:

- \- HP's version of DebitCredit (as documented in *HP 3000 Debit Credit Report*, December 1988)
- \- IBM's published AS/400 relative performance
- \- Discussions with the independent laboratory which conducted the DebitCredit tests of the AS/400 B30 under contract to HP

- IBM's AS/400 performance as reported in *Report of Arthur Andersen & Co. -- RAMP-C Benchmark*, dated June 1, 1989.
- Aberdeen's extensive experience in analyzing OLTP benchmarks.

IBM's proprietary benchmark, RAMP-C, rates the AS/400 B60 as having less than 10% more capacity than the HPPA model 950. Our opinion is that RAMP-C is not representative of how Stanley-Vidmar runs applications, and that the RAMP-C methodology used to measure the HPPA 950 is suspect. IBM recently engaged Arthur Andersen & Co. to audit IBM's RAMP-C benchmark tests of IBM, HP and Digital Equipment computers.

Note that after Arthur Andersen ran the IBM RAMP-C benchmarks it was so impressed with Hewlett-Packard that the partner in charge of the IBM RAMP-C benchmark study announced that Arthur Andersen was now OEMing Hewlett-Packard systems. In interviews with Aberdeen Group, Richard Stuckey, the partner-in-charge at Arthur Andersen, stated that the Hewlett-Packard running TurboImage was *the most impressive* of all the systems in IBM's tests.

IMPLICATIONS

- \- The AS/400 does poorly on the DebitCredit test for several reasons: . the inefficiency of journaling

  . difficulty in completing transactions within 1 second . difficulty supporting large numbers of users

- \- The model B60 offers about 25% less performance than Stanley-Vidmar's HP 3000 model 70. The top-of-the-line AS/400 model B70 approximates the HP 3000 model 70 in performance on the DebitCredit benchmark. There is no further growth path today in the AS/ 400 product line, and multiple-AS/400 clustering is not a reasonable growth alternative.

\- It is inappropriate to compare HPPA with the AS/400 based on IBM's RAMP-C benchmark.

CONCLUSIONS

\- The AS/400 product line has insufficient capacity to handle Stanley-Vidmar's current work load, and no room for future expansion.

The online transaction processing performance of the AS/400 models B10, B20, B35, B45, B50, B60 and B70 is depicted in Figure 2. For comparison purposes, we have included Figure 1 values for the HP 3000 48 and 70 computers now installed at Stanley-Vidmar.

The vertical axis shows DebitCredit throughput in transactions per second where 95% of all transactions complete in a second or less. The horizontal axis shows the IBM AS/400 models operating under the OS/400 native file system relational database and the SQL/400 relational database product.

VI. COMPARE THE EXISTING STANLEY-VIDMAR COMPUTERS WITH THE HPPA AND AS/400 PRODUCT FAMILIES.

FINDINGS

The OLTP performance of the HP and AS/400 machines discussed in questions four and five are summarized in Figures 3 and 4. Figure 3 shows DebitCredit performance using the fastest file system -- the HP TurboImage DBMS and the AS/400 native file system. Figure 4 shows the same HP and IBM AS/400 models with relational databases -- HP's Allbase SQL and IBM's SQL/400 interface.

IMPLICATIONS

- Hewlett-Packard's Precision Architecture with HP's database software outperforms IBM's AS/400.

CONCLUSIONS

- The AS/400 does not offer a performance upgrade from Stanley-Vidmar's present situation. Stanley-Vidmar would have to implement multiple AS/400s to meet the operational requirements of the organization that it plans to implement using Hewlett-Packard systems.

Figure 3 depicts both Hewlett-Packard 3000 and IBM AS/400 using the DebitCredit benchmark. Performance, as measured by transactions completed per second, is shown on the vertical axis. Transactions in this Figure are using the fastest available file system -- the native file system in the case of AS/400 and TurboImage in the case of HP.

The horizontal axis shows the IBM AS/400 models B10, B20, B35, B45, B50, B60 and B70 and the HP 3000 models 48, 70, 925LX, 935, 950, 955, and 960.

Figure 3 depicts both Hewlett-Packard 3000 and IBM AS/400 using the DebitCredit

benchmark. Performance, as measured by transactions completed per second, is shown on the vertical axis. Transactions in this Figure are using a relational database system -- the SQL/400 product in the case of AS/400 and Allbase SQL in the case of HP.

The horizontal axis shows the IBM AS/400 models B10, B20, B35, B45, B50, B60 and B70 and the HP 3000 models 48, 70, 925LX, 935, 950, 955, and 960. Note the HP models 48 and 70 now installed at Stanley-Vidmar do not support Allbase SQL.

VII. IF AN HPPA MACHINE IS RECOMMENDED, CHOOSE AND JUSTIFY THE MODEL.

FINDINGS

- The 935 offers a 70% capacity improvement over today's model 70, and has an attractive price/performance. However, the 935 cannot be internally expanded to higher models, meaning a box-swap -- removing the old machine and installing the new computer -- is necessary should the 935 capacity be exceeded.

- The 950 offers roughly 15% more capacity than the 935 -- 1 MIPS more according to HP. However, the 950 can be upgraded with a simple CPU-swap to the 955 and 960. The substantially higher initial cost of the 950 versus the 935 is offset by the ease of upward expandability. A CPU-swap usually takes minutes to accomplish while a box-swap can take hours or days of downtime.

- The 955 has approximately three times the performance of the model 70. The price of the 955 was recently reduced when the new top-end 960 was announced in September, 1989.

- A unit of capacity is least expensive when purchased up front; upgrades always cost more than initial capacity purchases.

IMPLICATIONS

- The 935 is appropriate only if Stanley-Vidmar's load projections remain static for several years or longer. Otherwise, Stanley-Vidmar will be penalized by the upgrade cost plus the expense/aggravation of a box-swap.

CONCLUSIONS

- The entry point for Stanley-Vidmar in the HPPA product line is the model 950. Stanley-Vidmar's growth projections may dictate the model 955 as the entry point if system utilization growth above 25% per annum is expected within the next two years.

VIII. IS MIPS A VALID METRIC FOR COMPARING MACHINES SUCH AS THE AS/400 AND HPPA?

FINDINGS

- MIPS -- Millions of Instructions Per Second -- does not measure system throughput in modern computers. At best, MIPS is an easily-misconstrued measure of CPU performance. Even CPU performance is clouded by what kind of MIPS are used or implied: Dhrystone MIPS for workstations and technical performance, IBM MIPS, VAX MIPS (e.g. VUPS), etc. [Note that most midrange MIPS ratings are based upon the use of a DEC VAX C compiler on a VAX 11/780 -- upgrades to DEC's C compiler change the whole midrange industry MIPS landscape at any point in time!]

- Online systems need to measure online, database-intensive performance from all the components of the system working together. MIPS does not do this.

- The AS/400 MIPS have never been fully revealed by IBM for the reason that they are confusing at best. Since the AS/400 uses one central processor and multiple I/O processors working in concert, there is no accepted method of defining MIPS in multiprocessor architectures such as the AS/400.

- Many commercial hardware and software suppliers have joined an independent transaction processing council in order to establish a common, non-proprietary benchmark for measuring online, database-intensive performance. This benchmark is based on an earlier benchmark called DebitCredit.

- Advances in DBMS software performance are totally unrelated to MIPS yet can provide users with significant systems enhancement.

- **The effectiveness of an organization is measured by ROI and PAT, not MIPS. The proper amount of MIPS in any organization is that amount that maximizes ROI and PBT.**

IMPLICATIONS

- Software performance typically improves over time. This changes the underlying "MIPS" rating because more work can be done with the faster software on the identical hardware configuration. Thus, even if MIPS were valid, the number would not be static.

- MIPS are a poor measure of Stanley-Vidmar's workload or requirements.

CONCLUSIONS

- Stanley-Vidmar should avoid MIPS-based comparisons as simplistic and totally unreliable.

- Stanley-Vidmar should use an OLTP benchmark like DebitCredit in order to compare systems on paper.

IX. COMMENT ON THE IDEA OF SALES DOLLARS/MIPS AS A VALID BUSINESS MEASUREMENT. SUGGEST OTHERS.

FINDINGS

- Aberdeen is unaware of organizations using sales dollars/MIPS as a means of determining how much computing power is enough. As discussed above, MIPS are notoriously suspect as a measure of computing power.

- Sales dollars/MIPS ignores the *total* costs of computing, such as operations, systems programmers, training, and productivity. For example, published studies show that comparable size machines from DEC and Unisys cost considerably less than their IBM System/370 counterparts. Sales/MIPS does not measure MIS costs, and it appears to penalize organizations which trade off ever-cheaper "MIPS" for labor, capital or other constrained resource.

- Aberdeen suggests the following potential ratios as better suited to determining MIS costs and efficiency:

- . MIS budget/revenue - varies from 1% to 7%
- . MIS capital budget/capital budget - 40% national average
- . Online terminals/employee - some companies exceed 1.0
- . Backlog months reduced/year - how fast are you catching up
- . Debugged Code Lines/Programmer/Year - ranges from 2K

with 3GL conventional tools to 2M with generators.

IMPLICATIONS

- The effective use of MIPS in an enterprise is regarded as positive. The highest-MIPS companies such as American Express, American Airlines, and Citicorp are usually seen as innovative, profitable, and aggressively competitive. They use MIPS as an important part of their competitive arsenal.

- The implication that MIPS can be arbitrarily tied to Revenue ignores the many tradeoffs organizations make in using computers.

- If insufficient MIPS in an organization leads to lower PBT due to the need for additional operational employees and if the ability to meet customer requirements is bottlenecked due to a lack of MIPS, the organization is being "penny-wise and dollar foolish."

CONCLUSIONS

- MIPS/Sales should not be used to judge Stanley-Vidmar, nor should the ratio play a role in the size of the selected computer.

- Quantitative PBT, ROI and customer satisfaction factors should be correlated with the qualitative judgement of how well information systems are being used in a enterprise for operational effectiveness.

X. WHATDON'TYOULIKEABOUTTHEAS/400ANDHOWDOESHPADDRESS THOSE CONCERNS?

FINDINGS

- The AS/400 operating system lacks maturity. Users report numerous bugs and horrible patch management problems. OS/400 has been in the field for about one year, so such immaturity is to be expected. However, the problems with OS/400 directly affect production shops:

. minor power failures lead to major database crashes. Users advise installation of an uninterruptable power supply

. the implementation of Journaling and Checksum options cuts throughput by up to 60% -- a steep price. However, the nature of the AS/400 single-level storage architecture places a premium on using data integrity schemes like Journaling and Checksum. Aberdeen strongly recommends using the OS/400 Journaling and Checksum features in order to avoid having to restore multiple disk drives after a system crash. Restoration can keep the machine unavailable to users for hours.

- Aberdeen direct experience and anecdotal trade press coverage indicate users are not happy with the performance and capacity of their configurations. These users believe the performance they are achieving is less than they expected.

We are unaware of OS maturity and performance problems on HPPA at this time. HP, as was widely reported, had difficulties getting out MPE XL release 1.0 in 1987. By 1989, any immaturity issues had been addressed, and users report that MPE XL is a stable, production platform. On the performance side, numerous users have migrated from performance-constrained models 68 and 70 to HPPA. They are generally pleased with HPPA performance and say the HPPA performance is what they expected.

- Many early investigations of AS/400 applications characterized them as a "mile wide and an inch deep" -- fine for data-in data-out applications of the 70s but not mature enough for complex applications of the 1990s.

IMPLICATIONS

- IBM OS/400 will reach stability and maturity with a yet-to-be-released 1990 version.

- Aberdeen believes the AS/400 performance problems, as they relate to IBM software, will not be remedied before 1991.

CONCLUSIONS

- Time will cure the AS/400 ills -- but time is the one commodity Stanley-Vidmar cannot afford given the present machine saturation problem.

XI. SOME PEOPLE ADVISE BUYING ONLY THE AMOUNT OF COMPUTING HORSEPOWER THAT YOU NEED TO SATISFY TODAY'S COMPUTING NEEDS, SINCE UNIT COSTS DECLINE OVER TIME. HOW FAR INTO THE FUTURE SHOULD A MID-SIZED MANUFACTURING FIRM WITH A SMALL MIS STAFF PLAN THEIR EQUIPMENT NEEDS?

FINDINGS

- The midrange market is different from the mainframe market where annual incremental upgrades are common.

- Aberdeen recommends purchasing capacity sufficient for needs out to two years beyond the date planned for first production.

- As discussed in question VII, choosing an expandable model is crucial to minimize conversion and box-swapping costs.

- **Aberdeen research shows OLTP sites similar to Stanley-Vidmar are adding capacity at the rate of 40% per year.**

- The migration to relational databases has caused MIS planners to underestimate capacity: SQL is still less machine-efficient than hierarchical databases; importantly, users are taking advantage of the relational model to generate more queries and reports.

IMPLICATIONS

- Micromanaging capacity when HP is the sole source of supply is pointless. An IBM mainframe site has competitive alternatives in Amdahl and NAS. Stanley-Vidmar, once committed, has no reasonable alternative to paying HP's upgrade prices. Therefore, Stanley- Vidmar should cut its best deal up front while it has the greatest leverage.

- Starting a new machine at 40%-50% of capacity, as projected for Stanley-Vidmar, is good, conservative management practice.

- Stanley-Vidmar user computing demand has been constrained by poor response time and known computer resource limitations. By increasing computer supply, demand will probably

grow faster than anticipated -- suggesting a conservative capacity purchase is warranted.

- As Stanley-Vidmar users become able to use Allbase SQL for queries and reports, MIS will have fewer EasyReporter reports to write.

CONCLUSIONS

- Stanley-Vidmar MIS should plan for two year's application growth volume after the new MRP system is installed and also factor in a 50% increase in *user* query/report demand over this same period.

SELECTING A MIDRANGE COMPUTER FOR
STANLEY-VIDMAR

A REPORT BY ABERDEEN GROUP, INC. OCTOBER, 1989

October 20, 1989

Mr. Timothy E. Walsh
Manager of Information Systems Stanley-Vidmar
Division of the Stanley Works 11 Grammes Road
Allentown, PA 18103

Dear Mr. Walsh:
Attached is the final report of the study we conducted on behalf of Stanley-Vidmar.

The report examines the business and technical issues surrounding the proposed acquisition by Stanley-Vidmar of an IBM AS/400 or Hewlett-Packard Precision Architecture computer system to replace an existing computer system.

As you requested, we have organized our report to answer eleven questions posed by Stanley-Vidmar to Aberdeen Group.

Should you or your colleagues have any questions about the content of the report or our conclusions, please do not hesitate to contact me.

Yours truly,

Peter S. Kastner Vice President

attachment

AI Metadata 2024

1. Concise Summary of the Findings in the Document

The document is a detailed pitch from Aberdeen Group to Stanley-Vidmar, evaluating the IBM AS/400 and Hewlett-Packard 3000 (HP 3000) systems. Aberdeen Group presents its qualifications, comparing its expertise to other consulting firms like Gartner Group. The document provides an in-depth analysis of both systems, highlighting the strengths and weaknesses of each. Aberdeen concludes that the HP 3000, particularly the HP Precision Architecture (HPPA) models, is a better fit for Stanley-Vidmar due to its maturity, performance, and lower risk compared to the AS/400.

2. Summary of How Computer Technology Buyers Should React to the Document

Computer technology buyers should consider Aberdeen Group's recommendation to opt for the HP 3000 over the IBM AS/400. The document argues that the HP 3000 offers superior performance, reliability, and a lower risk of implementation issues. Buyers should prioritize systems that align with their current expertise and operational requirements, and avoid high-risk transitions that could lead to increased costs and operational disruptions.

3. Analysis on the Technology Discussed and Its Relation to Mainstream Computing in 2024

The HP 3000 and IBM AS/400 were pivotal in the evolution of midrange computing, offering robust solutions for medium-sized enterprises. In 2024, the principles of reliability, performance, and scalability discussed in the document remain crucial. Modern computing continues to emphasize these aspects, with cloud computing and hybrid IT environments providing scalable and reliable solutions. The shift towards cloud-native architectures and microservices reflects the ongoing need for systems that can handle complex, high-volume transactions efficiently.

4. Analysis on How the Contents of the Document Played a Role in Impacting the History of the Companies Mentioned

The document highlights Aberdeen Group's role in influencing technology decisions for companies like Stanley-Vidmar. By recommending the HP 3000, Aberdeen contributed to Hewlett-Packard's dominance in the midrange market during the late 1980s and early 1990s. IBM's AS/400, despite its initial challenges, eventually became a cornerstone of IBM's midrange offerings, evolving into the IBM i series. The competitive analysis and recommendations provided by consulting firms like Aberdeen helped shape the strategic directions and market positions of these technology giants.

5. Market Acceleration Catalysts

- Increased demand for reliable and scalable midrange systems.

- Advancements in transaction processing and database management.

- The need for systems with lower implementation risks and higher user satisfaction.

- Competitive pressures driving innovation in midrange computing.

- The evolution of consulting services providing strategic technology recommendations.

6. Keywords

Aberdeen Group, IBM AS/400, HP 3000, midrange computing, transaction processing, system performance, technology consulting, market analysis, Hewlett-Packard, system upgrade.

Citations:

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